Finite Correspondences over Abelian Varieties and Bounded Denominators

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Abstract

Let a variety over a number field map quasi-finitely to an abelian variety. We prove that the values of any rational function on its rational points meet each finitely generated multiplicative subgroup in a finite set. The same conclusion holds whenever the function takes only finitely many values on each geometric fiber of a morphism to an abelian variety. This consequence of semiabelian Mordell–Lang extends the rational-map obstruction to finite algebraic correspondences. It implies omission of almost every term of every fixed affine geometric progression, and weak approximation for the complement. For generically finite morphisms, sufficiently late represented terms must come from the exceptional locus. An explicit smooth surface shows why this exception is necessary. We also exclude bounded-denominator sets containing all integers when the witness fibers are soluble unirational varieties, or form a smooth projective family of genus-one curves over an abelian base. The latter argument uses a divisor-class map to the relative Jacobian, Siegel's theorem, and Merel's uniform torsion theorem. These conclusions identify restrictions on constructions of Diophantine sets; they do not prove undefinability of a ring of integers in its field of fractions.

1 Definitions and the main statement

Throughout, k is a number field and $\overline{k}$ is an algebraic closure. Varieties are reduced separated schemes of finite type. A rational function is evaluated on a specified open domain on which it is a morphism to $\mathbb{A}^1$. Restricting that domain is permitted. For a morphism $\pi : V \to A$ and $a \in A(\overline{k})$, write V_a for its geometric fiber. A morphism is *quasi-finite* if it is of finite type and all its geometric fibers are finite.

Definition 1.1. A subset $T \subset k$ has *uniformly bounded denominators* if there is a positive integer M such that $MT \subseteq \mathcal{O}_k$. It has the *weak-approximation property* if its diagonal image is dense in $\prod_{v\in S} k_v$ for every finite set S of places of k.

Theorem 1.2. *Let A/k be an abelian variety, let $\pi : V \to A$ be a morphism, and let $h : U \to \mathbb{A}^1_k$ be the restriction of a rational function to an open subset $U \subseteq V$. Suppose that*

$$h\big(U(\overline{k}) \cap V_a(\overline{k})\big) \quad \text{is finite for every } a \in A(\overline{k}). \tag{1}$$

Then, for every finitely generated subgroup $\Lambda \subset k^*$, the set

$$h(U(k)) \cap \Lambda$$

is finite. In particular, the conclusion holds if π is quasi-finite.

The condition is about geometric fibers. Finiteness of the sets $U(k) \cap V_a(k)$ alone is not the stated hypothesis. Finite covers of arbitrary degree satisfy the hypothesis; no rational section or point-stability assertion is required.

For rational maps on abelian varieties, Corvaja–Zannier prove that finite unions of their rational value sets omit infinitely many integers and have a complement satisfying weak approximation [3, Theorems 1.1 and 1.10]. Theorem 1.2 permits outputs that are algebraic over the abelian coordinates, provided their geometric choices remain finite at every relevant point.

2 The constructible-image argument

We first record the arithmetic theorem in the form needed here. A semiabelian variety is an extension of an abelian variety by a torus.

Theorem 2.1 (Semiabelian Mordell–Lang)**.** *Let G be a semiabelian variety over a number field and let $\Gamma \subset G(\overline{k})$ be finitely generated. The Zariski closure of any subset of Γ is a finite union of translates of connected algebraic subgroups of G.*

This is a standard closure form of the semiabelian theorem [1, 2]. Indeed, if Z is the closure of the given subset, the usual statement applies to $Z \cap \Gamma$; that intersection is dense in Z. Taking closures of its finitely many subgroup cosets gives the assertion. One may equivalently use the integral-point form in [2, Theorem 2.2], after choosing one ring of S-integers containing the finitely many group generators and their inverses in a group model. Splitting a disconnected subgroup into its finitely many connected components gives the stated form. All subsequent subgroup arguments may be made over $\overline{k}$.

Lemma 2.2. *Every connected algebraic subgroup of $A \times \mathbb{G}_m$, where A is an abelian variety over an algebraically closed field of characteristic zero, is of one of the forms*

$$B \times \{1\}, \qquad B \times \mathbb{G}_m,$$

where B is an abelian subvariety of A.

Proof. Let H be such a subgroup, and let B be its image in A. The image of a homomorphism of algebraic groups is closed, so B is an abelian subvariety. The kernel of $H \to B$ is a subgroup of $\mathbb{G}_m$, hence is either finite or all of $\mathbb{G}_m$.

In the latter case each nonempty fiber contains every multiplicative coordinate, giving $H = B \times \mathbb{G}_m$. In the former case the surjective homomorphism with finite kernel is finite. Thus H is proper because B is proper. Every morphism from the connected proper variety H to $\mathbb{G}_m$ is constant. Its value at the identity is 1, so $H = B \times \{1\}$. □

Lemma 2.3. *Let $W \subset A \times \mathbb{G}_m$ be constructible, with finite geometric fibers under the first projection. If $\Lambda \subset k^*$ is finitely generated and*

$$\Sigma \subset W(k) \cap (A(k) \times \Lambda),$$

then the second coordinates of the points of Σ form a finite set.

Proof. The Mordell–Weil theorem makes $A(k) \times \Lambda$ finitely generated. Apply Theorem 2.1 to the closure $Z = \overline{\Sigma}$. It is essential here to close this actual arithmetic subset, rather than all of W.

After splitting components, every irreducible component of Z is a translate of one of the groups in Lemma 2.2. Suppose that one component is $(a+B) \times \mathbb{G}_m$. The subset Σ is dense in this component after the other irreducible components have been removed. Hence its constructible intersection with W is dense and contains a nonempty open subset of $(a + B) \times \mathbb{G}_m$. Choose a geometric point in that open subset. Its intersection with the vertical fiber through the point is a nonempty open subset of $\mathbb{G}_m$, and is therefore infinite. This contradicts the finite-fiber hypothesis on W.

Every component of Z consequently has a constant second coordinate. There are finitely many components, proving the assertion. Translates by torsion points and disconnected groups introduce only finitely many such components and cause no exception. □

Proof of Theorem 1.2. Restrict U to $h^{-1}(\mathbb{G}_m)$ and take its image W under (π, h). Chevalley's theorem makes W constructible [7, Tag 054K]. Hypothesis (1) says exactly that its geometric fibers over A are finite. Apply Lemma 2.3 to the image points coming from $U(k)$ with h-value in Λ. The lemma allows such a subset; it does not assume that every rational point of the constructible image has a rational lift to U. □

3 Arithmetic progressions and weak approximation

Corollary 3.1. *Let T be a finite union of value sets satisfying Theorem 1.2, together with a finite subset of k. For every $a, b \in k$ with $b \neq 0$, and every finitely generated $\Lambda \subset k^*$,*

$$T \cap (a + b\Lambda)$$

is finite. In particular, for every integer $q \geq 2$, all but finitely many terms $a + bq^n$, $n \geq 0$, are outside T.

Proof. Replace each output h by $(h - a)/b$. This affine change preserves finiteness of its values on each geometric fiber, so Theorem 1.2 applies. A finite union and a finite added set preserve finiteness. Take Λ generated by q for the last assertion. □

Corollary 3.2. *The complement $k \setminus T$ in Corollary 3.1 satisfies weak approximation.*

Proof. Fix neighborhoods at finitely many places. Weak approximation in k supplies a point $a \in k$ inside all of them. Choose a nonzero $b \in k$ sufficiently divisible at the prescribed finite places, so that adding any element of the same absolute values as b at those places keeps the point in the neighborhoods. Choose a rational prime q below none of these finite places.

The values $a + bq^{-n}$ are distinct. At each prescribed finite place, q is a unit, so the chosen divisibility of b works for every n. At every archimedean place, bq^{-n} tends to zero. Thus all sufficiently late terms lie in the prescribed product of neighborhoods. Only finitely many belong to T, by Corollary 3.1 applied to the group generated by q. The remaining terms prove the assertion. □

For $a, b \in \mathbb{Z}$, $b \neq 0$, the omitted values $a + b2^n$ lie in the integer arithmetic progression $a + b\mathbb{Z}$. Thus none of these constructions contains any complete integer arithmetic progression. In particular it cannot supply a set S with

$$\mathbb{Z} \subseteq S, \qquad MS \subseteq \mathcal{O}_k \tag{2}$$

for a fixed positive integer M. This obstruction concerns the first inclusion already; it needs no denominator hypothesis. The proof supplies no effective bound on the omitted exponents.

4 Algebraic outputs and exceptional loci

Proposition 4.1. *Let Y be a locally closed subvariety of an abelian variety A/k, let $c_0, \dots, c_d$ be regular functions on Y, and put*

$$C = \{a \in Y : c_0(a) = \cdots = c_d(a) = 0\}.$$

Suppose $\pi : V \to Y$ and a regular output $h : U \to \mathbb{A}^1$ satisfy

$$\sum_{j=0}^{d} c_j(\pi(v))h(v)^j = 0 \qquad (v \in U).$$

For each fixed $a, b \in k$, $b \neq 0$, and finitely generated $\Lambda \subset k^*$, only finitely many values in $a + b\Lambda$ have a witness with $\pi(v) \notin C$. Every other represented value in that set has all its witnesses above C. If the relation is monic in h, its full value set misses all but finitely many terms of every affine geometric progression.

Proof. Above $Y \setminus C$ the polynomial in the output is nonzero, so it has finitely many geometric roots. This gives (1) for the composite map to A, regardless of how many further witnesses lie above each point. Apply Corollary 3.1. A value outside that finite set cannot have any witness above $Y \setminus C$. In the monic case C is empty. □

Thus adding unrestricted auxiliary variables cannot repair a monic algebraic output relation over abelian coordinates. For a relation with varying leading coefficient, the common-zero locus of all coefficients must be checked: there the relation imposes no condition on the output.

Proposition 4.2. *Let V be integral and let $\pi : V \to A$ be generically finite onto its image. Let V_{qf} be its quasi-finite locus and $E = V \setminus V_{\mathrm{qf}}$. Then V_{qf} is a dense open subset. For a rational output h, all but finitely many represented values in any fixed set $a + b\Lambda$ have all their witnesses in E.*

Proof. The quasi-finite locus is open, and it contains the generic point when the extension of function fields is finite [7, Tag 02FW]. Restricting h to its intersection with that locus gives a value set satisfying Theorem 1.2. Apply Corollary 3.1. □

One can further stratify E into finitely many locally closed subvarieties and repeat the argument on each stratum whose map to A is quasi-finite. Such strata account for only finitely many values in a fixed affine multiplicative group. Remaining infinitely many represented terms must occur on strata with positive-dimensional geometric fibers. This is a restriction on the complete witness domain, not merely on its generic function field.

4.1 A smooth birational example with every field value

Proposition 4.3. *Over any characteristic-zero field k, put $E : y^2 = x^3 + 9$ and let $V \subset \mathbb{A}^4_k$, with coordinates (t, s, y, z), be given by*

$$y^2 = t^3 s^3 + 9, \qquad z^2 = s^3 + 9. \tag{3}$$

The surface V is smooth and geometrically irreducible. The map

$$\pi : V \longrightarrow E \times E, \qquad (t, s, y, z) \longmapsto ((ts, y), (s, z))$$

is birational onto the product of the affine elliptic charts. Its non-quasi-finite locus is exactly $s = 0$, and the regular output $h = t$ satisfies $h(V(k)) = k$.

Proof. On $s \neq 0$, the map is an isomorphism onto the open subset where the second abscissa is nonzero. Its inverse is

$$((x_1, y), (x_2, z)) \longmapsto (x_1/x_2, x_2, y, z).$$

That open subset is smooth and geometrically irreducible. When $s = 0$, equations (3) give $y, z \in \{3, -3\}$, with t arbitrary. The partial derivatives with respect to y, z form a diagonal submatrix with nonzero entries $2y, 2z$, so the surface is smooth there as well.

Every irreducible component of the two-equation locus in $\mathbb{A}^4$ has dimension at least two. Its part with $s = 0$ has dimension one, so every component meets $s \neq 0$. The irreducibility of that open subset therefore proves geometric irreducibility of V.

A fiber with $x_2 \neq 0$ is a singleton. If $x_2 = 0$ and $x_1 \neq 0$ it is empty. Above each of the four pairs with $x_1 = x_2 = 0$ the fiber is an affine line with coordinate t. These are exactly the non-quasi-finite fibers. In particular $(t, 0, 3, 3) \in V(k)$ for every $t \in k$, proving surjectivity. □

The equation $x_1 - tx_2 = 0$ is generically a degree-one equation for t, but both coefficients vanish at the four exceptional pairs. The surface is the affine ratio chart of the blow-up at these base points. Its exceptional lines explain the failure of an unguarded ratio formula. They introduce arbitrary field values even if the elliptic points themselves are restricted to a smaller field. They also have unbounded denominators at every finite place.

5 Unirational fibers and bounded denominators

Lemma 5.1. *Let $f \in k(T)$ be nonconstant. At infinitely many finite places of k, the valuations of $f(t)$ for $t \in k$ in any fixed nonempty Zariski-open domain are unbounded below. Consequently a nonconstant rational function on a k-unirational variety cannot have uniformly bounded denominators on its rational points in its domain.*

Proof. Write $f = F/G$ with coprime polynomials. If G is constant, then outside finitely many coefficient exceptions the leading term dominates when $v(t)$ is sufficiently negative, and $v(f(t))$ tends to $-\infty$.

Otherwise choose an irreducible factor H of G. After clearing denominators, its norm to $\mathbb{Q}$ is a nonconstant integer polynomial. Such a polynomial has prime divisors of its values at infinitely many rational primes. For completeness, if these primes formed a finite set P, choose an integer a where the polynomial Q is nonzero and put $D = Q(a)\prod_{p\in P} p$. Then $Q(a + DT)/Q(a)$ is a nonconstant integer polynomial congruent to 1 modulo every prime in P. A value of absolute value greater than one gives a prime divisor outside P, a contradiction.

Exclude the finitely many places coming from coefficient denominators, leading coefficients, and the nonzero resultants of H with H' and with F. Infinitely many remaining prime divisors of the norm give a place v where H has a simple residue root. Hensel's lemma gives a root $\alpha \in k_v$ of H, while $F(\alpha) \neq 0$. Thus f has a pole at α. Density of k in k_v supplies arguments approaching that pole, with values of arbitrarily negative valuation. A fixed finite set of excluded arguments does not alter the conclusion.

For the final assertion, compose a dominant rational parametrization by projective space with the given function. The result is nonconstant. Restricting to a suitable k-line gives a nonconstant one-variable rational function, defined away from finitely many points. Apply the first part. □

Lemma 5.2. *If $W \subset A \times \mathbb{A}^1$ is constructible, then the set*

$$B_0 = \{a \in A : W_a \text{ is finite geometrically}\}$$

is constructible. Moreover $W \cap (B_0 \times \mathbb{A}^1)$ is a constructible set with finite geometric fibers over A.

Proof. Decompose W into finitely many locally closed subvarieties. The dimension of the fibers of each projection has constructible level sets [7, Tag 05F9]. A constructible subset of a geometric affine line is finite exactly when its dimension is at most zero, including the empty case. Taking finite unions gives the assertion. □

Theorem 5.3. *Let $\pi : V \to A$ be a morphism to an abelian variety and let $h : U \to \mathbb{A}^1$ be a rational output on an open subset. Suppose that every fiber above $a \in A(k)$ meeting $U(k)$ is geometrically integral and k-unirational. If $T = h(U(k))$ has uniformly bounded denominators, then T has finite intersection with every affine set $a + b\Lambda$ as in Corollary 3.1. In particular it cannot contain $\mathbb{Z}$.*

Proof. Let W be the constructible image of (π, h). For a fiber contributing a rational witness, the restriction of h is a rational function on a geometrically integral unirational variety. If it

were nonconstant, Lemma 5.1 would contradict the bound on the denominators of T. Thus its entire geometric image on the domain is a single value.

Every actual rational image point therefore lies in $W \cap (B_0 \times \mathbb{A}^1)$, where B_0 is as in Lemma 5.2. This is constructible with finite geometric fibers. Apply Lemma 2.3, first to h and then to each affine transform $(h - a)/b$. □

Smooth conics and positive-dimensional smooth quadrics with a k-point are k-rational, by projection from that point. Theorem 5.3 therefore applies to families of such fibers over an abelian base. It also applies to unrestricted affine fibers. All equations imposed on auxiliary witnesses must be included in V before the hypothesis is checked.

Finitely many local preparation conditions do not by themselves remove the denominator problem in Lemma 5.1. Its proof supplies infinitely many places of unboundedness. Choose one outside the fixed places, and use weak approximation to approach its pole while retaining the prescribed nonempty open conditions elsewhere. A global denominator bound requires a further arithmetic restriction.

6 Genus-one families over an abelian base

We use two established arithmetic facts. Siegel's theorem implies that for a smooth projective curve C/k of positive genus and a nonconstant rational function f on C, only finitely many points in the domain have f-values in a fixed ring of S-integers [2, Section 1]. To see this formulation, use the finite map $C \to \mathbb{P}^1$ defined by f. After enlarging S, its inverse image of $\mathbb{A}^1$ has a model finite over $\mathbb{A}^1_{\mathcal{O}_k[S^{-1}]}$; a point with integral f-value is integral in that model by properness of the finite map. Siegel's theorem applies to this affine positive-genus curve.

Merel's uniform torsion theorem supplies an integer $N = N(k) > 0$ that annihilates the rational torsion subgroup of every elliptic curve over k [4, Introductory Corollary]. One can take a common multiple of all positive integers below the uniform bound for the order of a torsion point.

We first describe how a relative divisor replaces the choice of an origin. For a smooth projective family of geometrically connected curves, the identity component of the relative Picard scheme is its relative Jacobian; for genus one it is an elliptic scheme [5, Section 9.4, Theorem 1].

Lemma 6.1. *Let $\mathcal{C} \to B$ be a smooth projective family of geometrically connected genus-one curves, and let $\mathcal{L}$ be a line bundle of constant positive degree d on its geometric fibers. Write $\mathcal{J} = \mathrm{Pic}^0_{\mathcal{C}/B}$ for its relative Jacobian. The formula*

$$\alpha_{\mathcal{L}} : \mathcal{C} \longrightarrow \mathcal{J}, \qquad P \longmapsto [\mathcal{O}_{\mathcal{C}_b}(dP) \otimes \mathcal{L}_b^{-1}]$$

defines a finite morphism whose degree on every geometric fiber is d^2.

Proof. On $\mathcal{C} \times_B \mathcal{C}$, the diagonal is a relative effective Cartier divisor. The line bundle $\mathcal{O}(d\Delta) \otimes \mathrm{pr}_2^* \mathcal{L}^{-1}$ has degree zero on the fibers over the first factor. It therefore defines the displayed morphism to the relative Picard scheme.

On a geometric fiber choose any origin O. Identifying that curve with its Jacobian, the morphism becomes

$$P \longmapsto [d](P - O) - [\mathcal{L}_b \otimes \mathcal{O}(-dO)].$$

This is a translate of multiplication by d, so it is finite of degree d^2; this divisor-class construction is also described in [6, Section 1]. The relative morphism is proper, since $\mathcal{C}$ is proper over B and $\mathcal{J}$ is separated over B. Its geometric fibers are finite, so it is finite. □

Theorem 6.2. *Let B be a locally closed subvariety of an abelian variety A/k, and let $\mathcal{C} \to B$ be a smooth proper family of geometrically connected genus-one curves, equipped with a relatively ample line bundle. Let $h : U \to \mathbb{A}^1$ be a rational function on its total space, on an open domain U. If $T = h(U(k))$ has uniformly bounded denominators, then T has finite intersection with every set $a + b\Lambda$ of Corollary 3.1. Hence T cannot contain $\mathbb{Z}$. No section of $\mathcal{C} \to B$ is required.*

Proof. The relatively ample bundle makes the proper morphism projective. Let W be the constructible image of U in $A \times \mathbb{A}^1$. Separate its finite-fiber locus using Lemma 5.2. Outputs above that locus satisfy Lemma 2.3. This includes empty domains on a fiber. On a nonempty domain in a geometric genus-one fiber, the image is finite exactly when the induced function is constant.

Consider a remaining fiber over $b \in B(k)$ that contributes a rational witness. The function induced by h on $\mathcal{C}_b$ is nonconstant, and its domain omits only finitely many geometric points. The fixed denominator bound and Siegel's theorem make $U(k) \cap \mathcal{C}_b(k)$ finite. This fiber has a rational point, so choosing one identifies its rational points with those of its Jacobian $\mathcal{J}_b$. If $\mathcal{J}_b(k)$ had positive rank, the cofinite open part would have infinitely many rational points, a contradiction. Thus $\mathcal{J}_b(k)$ consists entirely of torsion points.

Let $\mathcal{L}$ be the given relatively ample bundle. Its degree is a positive locally constant integer; splitting the finitely many connected components of B, assume it equals d. Choose $N = N(k)$ as above and use Lemma 6.1. For every rational witness P over a nonconstant contributing fiber, $\alpha_{\mathcal{L}}(P)$ belongs to $\mathcal{J}_b(k)$ and hence to $\mathcal{J}_b[N]$. All such witnesses therefore lie in

$$\mathcal{F} = \alpha_{\mathcal{L}}^{-1}(\mathcal{J}[N]).$$

Both $\alpha_{\mathcal{L}}$ and $\mathcal{J}[N] \to B$ are finite, so $\mathcal{F} \to B$ is finite. Its composite map to A is quasi-finite. Apply Theorem 1.2 to h on the reduced scheme underlying $\mathcal{F} \cap U$. Together with the finite-output part, this covers all actual rational witnesses. Apply the same argument to $(h - a)/b$ for the affine assertion; its denominators remain uniformly bounded after multiplication by a fixed integer. □

For an elliptic scheme with zero section O, take $\mathcal{L} = \mathcal{O}(O)$ and $d = 1$; then $\alpha_{\mathcal{L}}$ is the identity after the usual identification with its Jacobian. For a genus-one torsor the divisor-class map avoids making a varying arithmetic choice of an origin. The finite subscheme $\mathcal{F}$ replaces the torsion subscheme of an elliptic scheme.

The theorem also applies to any open part of such a smooth projective family, and to finitely many such families on locally closed strata of the base. Extra witness equations must be included in the source before applying it. It does not assert that every family with finitely many rational points on each fiber has a finite geometric subscheme containing them. In particular the argument does not extend to higher-genus families merely by applying Faltings's theorem: their Jacobians may have positive rank even when the curves have only finitely many rational points.

7 Consequences for defining rings of integers

Recall that a subset of k^m is *Diophantine over k* if it is the image under a coordinate projection of the common zero set in k^{m+n} of finitely many polynomials with coefficients in k, for some $n \geq 0$.

For a number field k, a Diophantine set satisfying (2) would suffice to define $\mathcal{O}_k$ in k. Indeed

$$S' = \bigcup_{j=0}^{M-1} (j + MS)$$

satisfies $\mathbb{Z} \subseteq S' \subseteq \mathcal{O}_k$. For an integral basis $\omega_1, \ldots, \omega_d$ one then has

$$\mathcal{O}_k = \omega_1 S' + \cdots + \omega_d S'.$$

Both finite unions and these sums preserve Diophantine definability.

The preceding theorems exclude this strategy when all output values come from the finite correspondences of Theorem 1.2. Under the denominator hypothesis they also exclude the unirational and genus-one families specified above. These obstructions hold over every number field, independently of its real or complex embeddings. For a generically finite description, the exceptional loci must be included in the analysis: they can contain every represented input, as Proposition 4.3 demonstrates.

No assertion here concerns arbitrary existential varieties. In particular, the results do not produce a number field whose ring of integers is proved not to be Diophantine in the field. They identify where a construction must introduce additional arithmetic: through output-varying positive-dimensional fibers that survive the stated denominator obstructions, or through a different source of witnesses.

8 Exact algebraic verification

The standard-library program `check_exceptional_correspondence.py` records the exact identities in Proposition 4.3: the two curve equations, the inverse on $s \neq 0$, all four exceptional lines, and the nonzero boundary Jacobian minors. It also records the symbolic family $(t, 0, 3, 3)$, whose output is unrestricted. These finite checks verify the displayed example. The general finiteness and denominator theorems are established by the proofs and the cited arithmetic inputs, not by numerical sampling.

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